

Machine Learning Engineering professional targeting Senior/Lead roles, with extensive experience building scalable, production-grade AI/ML systems and enterprise solutions. Expertise across Machine Learning, Deep Learning, NLP, LLMs, ML pipelines, and deployment, with an architecture-oriented approach to delivering high-performance solutions as a senior technical individual contributor.

PROFILE SUMMARY

- Senior Machine Learning Engineer with 9+ years of experience in building and productionizing enterprise-scale AI/ML solutions, with end-to-end exposure across model development, evaluation, optimization, deployment, and production operations.
- Proven experience in delivering large-scale intelligent automation solutions, including a 643-class mortgage solution processing 1.7M requests/day with 99.99% success and production processing of 2.8M+ pages across 49 encrypted lender sources.
- Strong background in Deep Learning, Computer Vision, NLP, Multimodal Learning, and Generative AI, with experience developing and optimizing models for complex enterprise use cases and achieving up to 0.975 weighted F1 through distributed and multi-GPU training.
- Practical experience in building LLM and RAG-based applications, including retrieval, reranking, structured generation, and model adaptation, with exposure to modern GenAI and Agentic AI architectures.
- Experienced in engineering scalable ML pipelines and production platforms, with a focus on performance optimization, deployment automation, monitoring, reliability, and operational efficiency, taking solutions from experimentation through production.
- Architecture-oriented technical individual contributor experienced in contributing to ML system design, production reliability, incident RCA, cost optimization, and release governance, targeting Senior/Lead Machine Learning Engineering roles focused on solving complex, large-scale ML challenges.

WORK EXPERIENCE

ICE DATA SERVICES | Senior ML Engineer | Hyderabad | Nov 2023 – Present

Highlights:

- Enterprise ML Scale: Supported a mortgage document classification platform processing 1.7M requests/day and 45M+ requests/month with 99.99% success rate and 1.8-second p95 latency.
- Business Transformation: Reduced document verification turnaround from ~4 days to 4–5 hours through production ML automation.
- Model Excellence: Delivered 0.975 weighted F1 on 528K+ pages, outperforming the existing OCR-based production ensemble.
- Accuracy Recovery: Restored production F1 from 0.62 to 0.92 across 946,355 pages by resolving a train-to-benchmark taxonomy mismatch and improving prediction calibration.
- Cost Optimization: Achieved approximately 20% lower hourly infrastructure cost through model and deployment optimization.
- Production Leadership: Delivered 4 consecutive ML releases and scaled production infrastructure from 20 to 118 SageMaker instances while maintaining platform reliability.

Key Result Areas:

- Own the end-to-end ML lifecycle across model development, evaluation, optimization, deployment, monitoring, and production operations for 643 document classes and 49 encrypted lender sources.
- Develop and scale multimodal classification architectures using ResNet50, CNN, BiLSTM, and custom multi-head-attention Transformers; distribute training across 20 A10G GPUs using Horovod, resolving host-memory OOM and distributed all-reduce failures and achieving 0.959–0.972 weighted F1 across five lender benchmarks.
- Develop OCR-free classification models using Donut/Swin, fine-tuning 75M parameters with PyTorch Lightning DDP across 20 L40S GPUs using BF16; export models to ONNX after benchmarking 9 SageMaker instance types.
- Recover a production accuracy regression by fixing the train-to-benchmark taxonomy mismatch that took F1 from 0.62 to 0.92; establish that 64% of residual errors were valid predictions discarded by confidence thresholds, leading to calibrated per-class selective prediction across all 643 classes.

CORE COMPETENCIES

- Machine Learning & Multimodal Deep Learning
- Computer Vision & OCR-Free AI
- NLP & Information Extraction
- LLM Engineering & Generative AI
- RAG & Enterprise Retrieval Systems
- Transformer Architecture & Model Fine-Tuning
- Distributed ML & Large-Scale Model Training
- MLOps & Production AI Engineering
- AWS SageMaker & Cloud AI Engineering
- Model Optimization, Inference & AI Reliability
- AI Solution Architecture & Technical Leadership

TECHNICAL SKILLS

- Machine Learning & Deep Learning: PyTorch | PyTorch Lightning | TensorFlow | Keras | Hugging Face Transformers | CNN | ResNet50 | BiLSTM | Swin Transformer | Vision Transformer (ViT) | Donut | Multimodal Fusion | Named Entity Recognition (NER) | LoRA | QLoRA | PEFT | Supervised Fine-Tuning (SFT) | Instruction Tuning | RLHF/DPO | Mixed Precision (FP16/BF16)
- Generative AI, LLM & RAG: Large Language Models (LLMs) | Generative AI | Retrieval-Augmented Generation (RAG) | FAISS | Dense Retrieval | Okapi BM25 | Hybrid Retrieval | Reciprocal Rank Fusion (RRF) | Cross-Encoder Reranking | Structured Outputs | JSON Schema | Llama 3 | Anthropic Claude API | Selective Prediction | Confidence Calibration
- Computer Vision & Document AI: Computer Vision | OCR/OCR-Free Processing | OpenCV | Tesseract | AWS Textract | TensorFlow Object Detection | Image Preprocessing | Document Classification | Information Extraction | Object Detection | Signature Verification | Document Deduplication
- Distributed ML & Data Engineering: Distributed Data Parallel (DDP) | Horovod | TorchDistributor | NCCL | Multi-GPU Training | Apache Spark | PySpark | Databricks
- MLOps, Deployment & Model Serving: AWS SageMaker | MLflow | Model Registry | Real-Time Model Endpoints | Auto Scaling | TensorFlow Serving | ONNX | Docker | Jenkins CI/CD | AWS CloudWatch | Model Optimization | Production Monitoring
- Programming, Databases & Platforms: Python | SQL | Git | Linux | Bash

PERSISTENT SYSTEMS | Senior ML Engineer | Client: ICE Data Services | Hyderabad | Aug 2021 – Nov 2023

- Engineered the foundational data ingestion and OCR pipeline supporting ICE's production document classification ecosystem, processing encrypted lender image sets, extracting text at scale with Tesseract, and consolidating 27+ source schemas into a unified labelled corpus using PySpark UDFs.
- Developed an OpenCV-based document preprocessing pipeline incorporating thresholding, denoising, skew correction, and image normalization, improving OCR quality and downstream model performance across diverse lender inputs.
- Built the first multimodal CNN + BiLSTM training pipeline on Databricks, including the BiLSTM encoder-decoder responsible for generating sequence representations consumed by downstream text-processing branches.
- Established the initial AWS SageMaker deployment and monitoring architecture, together with MLflow experiment tracking, providing the foundation for the subsequent production MLOps and Jenkins-to-SageMaker release pipeline.
- Supported the evolution of the initial ML architecture into the production-scale 643-class document classification ensemble.

ZYCLYX | Data Scientist | Client: Al Rajhi Bank, Riyadh | Riyadh | Aug 2019 – Mar 2021

- Designed and built the bank's document digitization platform as its first ML Engineer, processing approximately 100K bilingual Arabic/English documents through an end-to-end classification, detection, OCR, and entity-extraction pipeline.
- Developed a ResNet50 document classifier covering ~65 document types, achieving approximately 88% classification accuracy, followed by TensorFlow Object Detection for field localization, region cropping, Tesseract OCR, and GRU-based NER for extracting names, countries, identification numbers, and other key entities.
- Engineered a deep-learning signature verification model to identify potentially forged signatures and strengthen document-level fraud detection.
- Diagnosed the primary upstream source of OCR errors by auditing scanning configurations across major bank branches; identified inconsistent scanner settings causing colour distortion, rotation, and skew, and established a standardized configuration subsequently mandated bank-wide through an official bank circular.
- Integrated ML extraction services into the bank's BluePrism RPA workflow with a human-in-the-loop validation layer, enabling page-level and field-level verification while feeding corrections back as training signals.
- Productionized the platform on-premise using Flask and Gunicorn microservices, ensuring sensitive documents remained within the bank's network.
- Mentored 5+ engineers as the ML team expanded, accelerating their capabilities across OCR, document processing, model development, and training workflow.

VISSINDIA | Python & Computer Vision Engineer | Hyderabad | Feb 2017 – Aug 2019

- Developed real-time computer-vision solutions for identity verification and multi-object tracking using Haar Cascades, HOG, CNNs, SIFT, SURF, FLANN, TensorFlow Object Detection, and OpenCV.
- Built image and video preprocessing pipelines to improve the quality and consistency of noisy visual inputs before model inference.
- Developed detection, feature-matching, and tracking workflow for real-time video analytics and identity-verification applications.



MAJOR PROJECTS

Rulebook-RAG | Retrieval-Grounded Document Classification | Personal Project | 2026

- Built a 36-class RAG classifier without labelled training data or model retraining, using a dynamically indexed rulebook.
- Enabled zero-retraining class expansion, achieving 0.83 accuracy on 5 newly added classes.
- Implemented BM25 + dense retrieval + RRF, improving accuracy by 5.2 pts; cross-encoder reranking added another 3.9 pts.
- Delivered citation-grounded predictions with live rule validation, achieving 0 invalid citations across 775 predictions.

Document Deduplication Pipeline | ICE Data Services

- Built near-duplicate detection across a 7.1M-page mortgage corpus using Donut/Swin embeddings across 40 GPU nodes.
- Implemented two-tier FAISS cosine search at page and document levels using mean-pooled embeddings.
- Optimized exact retrieval through label/page-count grouping and transitive Union-Find clustering.
- Identified 67% of pages and 69% of documents as near-duplicates for large-scale corpus rationalization.



EDUCATION

- B.Tech. - Computer Science & Engineering: TKR Engineering College, JNTU, Hyderabad | 2016